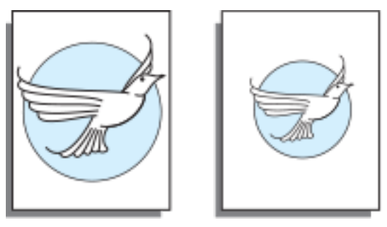
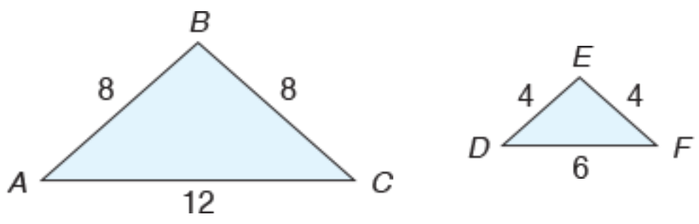
Lessons

Missing Sides in Similar Figures

Similar figures have the same shape but are different sizes. Imagine using a copy machine to reduce a figure you have drawn. The figure on the copy would be in a smaller scale than the original, but the shape would be the same.

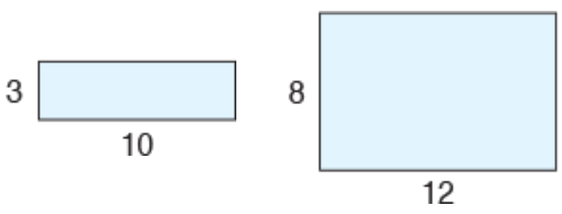


The triangles below are similar because the ratios of the lengths of corresponding sides are equal. Each side in the second triangle is half as long as the first.



ABC ~ *DEF* The symbol ~ means "is similar to."

The rectangles shown are not similar.



You can use proportion to solve problems about similar figures.

Example 1

Hide

The plots of land shown here are similar. Find the width of the smaller plot.

Step 1

Set up a proportion with corresponding sides.

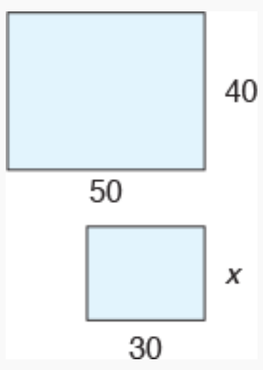
$$\frac{\text{length}}{\text{width}}\ \frac{50}{40} = \frac{30}{x}$$

Step 2

Cross multiply and solve for x.

$50x = 1{,}200$

$x = 24$



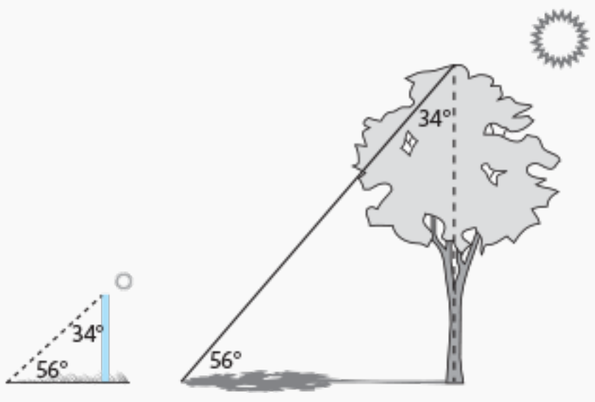
The width of the smaller plot is 24.

Two triangles are similar if their corresponding angles are equal. When the sun casts the shadow of an object that is perpendicular to the ground, a right triangle is formed. The angle of the sun will be the same for any two objects, creating equal corresponding angles. You can find the height of a tall object by using a smaller object and comparing similar triangles.

Example 2

Hide

A street sign 6 feet high casts a shadow of 4 feet. At the same time, a nearby tree casts a 10-foot shadow. How tall is the tree?



Step 1

Set up a proportion with corresponding sides.

$$\frac{\text{height}}{\text{shadow}}\ \frac{6}{4} = \frac{x}{10}$$

Step 2

Cross multiply and solve for x.

$4x = 60$

$x = 15$

The height of the tree is **15 feet.**

Category

Lesson 6: Problem Solving with Triangles

Related Questions:

Practice questions taken related to this topic

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